

What is Java String?

- In Java, a String represents a sequence of characters.
- Examples: "Hello", "Java", "Welcome to my app".
- Strings are objects of the String class, not primitive types.
- They offer built-in methods for working with text (length, join, uppercase, etc.).
- Strings are immutable – once created, their value cannot be changed.
- Any modification creates a new String in memory.
- This makes Strings safe, especially in multi-threaded environments.
- Strings are used everywhere – printing, input handling, file I/O, web apps.
- Learning Strings is a foundational step in mastering Java.

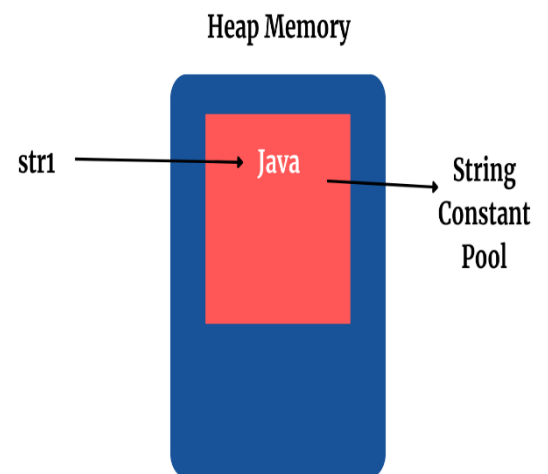
Why Strings Are Immutable in Java

- Immutable means **unchangeable** — the object's state cannot be modified after creation.
- **Immutable objects** are safe to use across multiple threads — they remain consistent.
- In Java, **String is an object** — not a primitive type.
- Once a String is created, **its value cannot be changed**.
- Any modification creates a **new String object** behind the scenes.
- The original string **remains unchanged**.
- The String class is **final**, so its behavior can't be overridden.
- This makes Strings **secure, reliable, and thread-safe**.

String Constant Pool in Java

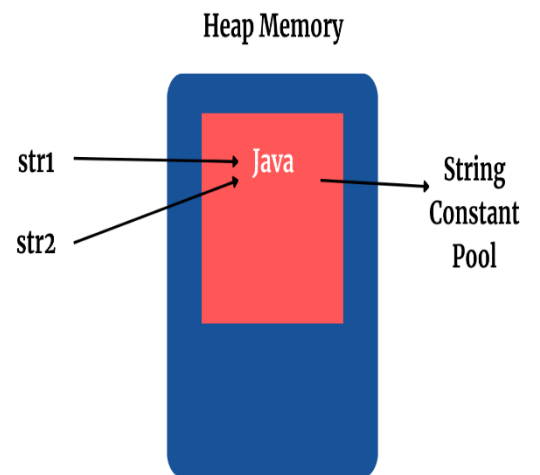
In Java, whenever we create a string using double quotes — like "Java" — it gets stored in a special memory area called the String Constant Pool. This pool is part of the heap memory, and it's designed to save space by avoiding duplicate strings.

```
String str1 = "Java";
```



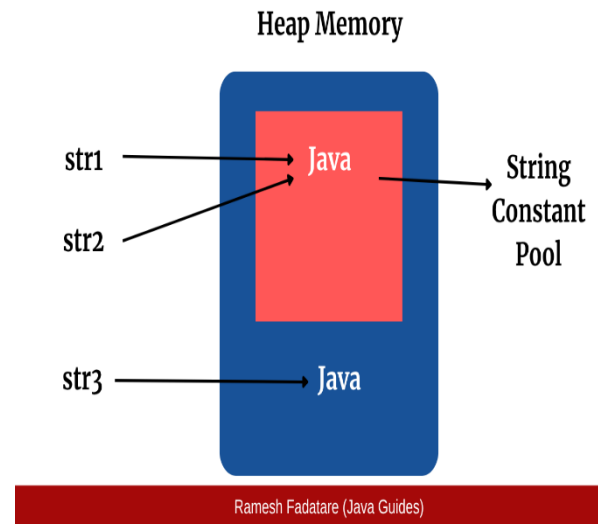
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```
String str2 = "Java";
```



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```
String str3 = new String("Java");
```



String Comparison

The `==` operator checks if two string references point to the exact same memory location — in other words, it checks if both variables refer to the same object.

But the `equals()` method checks whether the contents of two strings are the same, even if they're different objects.

This means two strings can contain the same characters, like "Java" and "Java", but still be different objects — and in that case, `==` would return false, but `equals()` would return true.

How to find the length of a string in Java

In Java, when we talk about the length of a string, we mean **how many characters are inside it**. That includes letters, numbers, spaces — every single character.

So, if we have the string "Java", the length is 4 — because it has J, a, v, and a.

And to find the length of any string in Java, we use a method called **`length()`**.